Technology Stack

Date	27-06-2026
Team ID	SWTID-2026-3845
Project Name	SMART LENDER
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1	**Programming Language**	*Python*	Rich ecosystem of ML/data-science libraries and easy integration with web frameworks.

2	**Web Framework**	*Flask*	Lightweight, easy to set up, and well-suited for serving ML models via a simple web interface.
3	**Data Processing**	*Pandas, NumPy*	Efficient handling, cleaning, and transformation of structured applicant data.
4	**Machine Learning Libraries**	*Scikit-learn, XGBoost*	Provides Decision Tree, Random Forest, KNN, and XGBoost classifiers; XGBoost selected as the best-performing model
5	**Model Serialization**	*Pickle / Joblib*	Saves the trained XGBoost model so it can be loaded quickly for real-time inference without retraining.

6	**Cloud Deployment Platform**	*Render*	Hosts the Flask application, enabling scalable, remote access for banks and financial institutions.